

Why the measured bias ratio runs above r^2 at large r

The r^2 prediction was a leading-order limit

In `sigma_mismatch_full_derivation.pdf` we wrote

$$\frac{|\delta h_w|}{|\delta h_z|} \simeq r^2, \quad (1)$$

but this used the approximation $r^2 \ll R^2$ in the curvature denominators. Keeping *all* terms gives the exact Newton-step ratio.

The exact Newton-step ratio

The two Cases differ only in their second-derivative denominators:

$$\text{Case A (mismatch } \sigma_w): \quad \text{NLL}''(\nu_t) = \frac{V_z''}{\sigma_z^2} + r^2 \frac{V_w''}{\sigma_w^2} \simeq 2t^2(R^2 - r^2), \quad (2)$$

$$\text{Case B (mismatch } \sigma_z): \quad \text{NLL}''(\nu_t) = r^2 \frac{V_z''}{\sigma_z^2} + \frac{V_w''}{\sigma_w^2} \simeq 2t^2(R^2 r^2 - 1). \quad (3)$$

The two gradients (numerators) have equal magnitude $\propto (r^2 - 1)$, so:

$$\boxed{\frac{|\delta h_w|}{|\delta h_z|} = \frac{R^2 r^2 - 1}{R^2 - r^2}.} \quad (4)$$

Expanding the correction

Factor out r^2 :

$$\frac{|\delta h_w|}{|\delta h_z|} = r^2 \cdot \underbrace{\frac{1 - 1/(R^2 r^2)}{1 - r^2/R^2}}_{\text{correction factor}} \approx r^2 \cdot \frac{1}{1 - r^2/R^2}. \quad (5)$$

The correction factor is ≥ 1 and grows monotonically with r . For $r \ll R$ it is ≈ 1 , recovering the r^2 scaling. As $r \rightarrow R$ it diverges — at $r = R$ the Case-A curvature vanishes and the bias truly runs away.

Numerical example with $R_{\text{eff}} = 20$

r	r^2/R^2	correction	predicted ratio	pure r^2
1.5	0.006	1.006	2.27	2.25
2.0	0.010	1.010	4.04	4.00
3.0	0.022	1.023	9.21	9.00
5.0	0.063	1.067	26.67	25.00
10	0.250	1.333	133.3	100.0

At $r = 10$ with $R_{\text{eff}} \sim 20$, the measured ratio sits roughly 33% above the r^2 line — exactly the trend you observed.

Why this happens physically

In Case A, shrinking $\sigma_{w,\text{fit}}$ by a factor r multiplies the *negative* curvature of the V_w -direction by r^2 . That negative curvature cancels against the positive V_z -curvature. The total curvature is $\propto (R^2 - r^2)$ — *decreasing* with r . As the valley flattens, the optimum slides further before the (now weaker) restoring force stops it. So $|\delta h_w|$ grows *faster* than r^2 .

In Case B, both curvatures contribute with the same sign — the valley only stiffens as r grows. The bias $|\delta h_z|$ decreases with r at the rate the leading-order formula predicts. No correction.

How to extract R_{eff} from the data

Fit the measured ratios to (4) with R_{eff} as a single free parameter:

$$\text{ratio}(r; R_{\text{eff}}) = \frac{R_{\text{eff}}^2 r^2 - 1}{R_{\text{eff}}^2 - r^2}. \quad (6)$$

A clean fit with a physically sensible R_{eff} value (consistent with $\sigma_w/(\nu_t \sigma_z)$ for the simulation) confirms that the deviation from r^2 is the predicted curvature-cancellation effect, not anharmonicity or finite- N noise.

Summary

- r^2 is the leading-order prediction, valid for $r \ll R$.
- The exact ratio is $\frac{R^2 r^2 - 1}{R^2 - r^2}$, which always lies *above* r^2 and diverges as $r \rightarrow R$.
- Your observed up-curve at $r = 10$ is the precursor of the $r = R$ runaway.
- A one-parameter fit for R_{eff} on the measured ratios will confirm the mechanism quantitatively.